

FIT3080 Applied Class on Problem Solving as Search: Irrevocable and Adversarial Search

Exercise 1: Irrevocable Search Algorithms

- (a) What happens in Simulated Annealing if the temperature remains at ∞ at all times? What happens if the temperature starts at 0?
- (b) A genetic algorithm is to be used to evolve a binary string of length n containing only 1s. The initial population is a randomly generated set of binary strings of length n .
 - i. Propose a suitable fitness function for this problem.
 - ii. Calculate the probability of selecting each of the following strings according to your function.
 00110001
 01011101
 11101111

Exercise 2: Game Graphs

The game *nim* is played as follows: Two players alternate in removing one, two or three coins from a stack initially containing 5 coins. The player who picks up the last coin loses. Show by drawing the game graph, that the player who has the second move can always win. Can you think of a simple characterization of the winning strategy?

Exercise 3: Minimax and α - β Search

Consider a hypothetical game with branching factor 2. It is MAX's turn to play. They are able to evaluate a position 4 steps in advance. The following is a list of the values of the positions at the bottom of the game tree, should they ever need to be evaluated:

node #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
e(n)	3	9	1	8	0	13	6	20	2	4	7	5	11	14	8	13

- (a) Draw a step-by-step game tree, as evaluated by MAX using the minimax procedure with DFS-style exploration.
- (b) Draw a step-by-step game tree, as evaluated by MAX using the α - β procedure with DFS- or Backtrack-style exploration, so that only the expanded nodes appear in your diagram. Indicate clearly the α and β value of each node.
- (c) How many α cut-offs and β cut-offs have been performed?
- (d) What is the best move for MAX and what is its backed-up value?

Exercise 4: Minimax Algorithm (Optional, Python coding)

Consider the complete game tree for Exercise 3. Using the outcomes for the 16 terminal nodes in `lab4_minimax_template.py` on Moodle, implement the minimax algorithm to find the first move if playing as Max.